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CIS5500 Group 13
Final Project Milestone 4

Overview:
The API supports a housing affordability web application that enables users to explore housing values, income, and affordability across U.S. regions.
There are 3 tables: region, acs_income, and zhvi_value.
All routes are designed based on the analytical SQL queries defined in Milestone 3. Since the application is read-only and does not modify data, all routes use the GET method.

The response will be as below as an example.
Successful:
```
{
  "results": ~
}
```
Failed:
```
{
  "error": "Invalid request"
}
```

Route Specifications:

Route 1:
● Retrieves the total number of regions in the dataset. This corresponds to Milestone 3 Query 1
Description: Counts the total number of regions in the dataset.
```
SELECT COUNT(*) AS total_regions
FROM region;
```

● Request Path: /regions/count
● No request parameters
● No query parameters
● Response parameters: total_regions as integer: total number of rows in the region table

```
{
 "total_regions": x
}
```

Route 2:
● Retrieves the average household income across all regions for each year. This corresponds to Milestone 3 Query 2.
Description: Computes the average median household income across all regions for each year.

SELECT year, AVG(household_income) AS avg_income
FROM acs_income
GROUP BY year
ORDER BY year;

- Request path: /income/average-by-year
- No request parameters
- No query parameters
- Response parameters: year as integer: ACS year
  avg_income as float; average household income across all regions in that year

Route 3:

- Retrieves the most recent home value available for each region. This corresponds to Milestone 3 Query 3 below.

Description: Retrieves the most recent home value available for each region.

SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT MAX(date) FROM zhvi_value);

- Request path: /home-values/latest
- Request parameters, types, and descriptions: None.
- Query parameters: limit query as integer: optional maximum number of rows to return
  offset query as integer: optional number of rows to skip for pagination
- Response parameters: region_id as integer: region identifier
  value as float: most recent home value for the region

Route 4:

- Retrieves regions ranked by latest home value. This corresponds to Milestone 3 Query 4.
- Request path: /rankings/home-values
- Request parameters, types, and descriptions: None.
- Query parameters (parameters used in the SQL / NoSQL query):
  limit, query as integer: number of rows to return, default 25
  state_name, query as string: filter results to a specific state
  sort: as desc
- Response parameters, types, and descriptions:
  region_id as integer: region identifier
  region_name as string: region name
  state_name as string: state abbreviation or state name
  value as float: latest home value

Route 5:

- Retrieves regions ranked by most recent household income. This corresponds to Milestone 3 Query 5.

● GET /rankings/income
● Request parameters, types, and descriptions: None.
● Query parameters (parameters used in the SQL / NoSQL query):
    limit, query as integer: number of rows to return
    state_name, query as string: filter results to a specific state
    sort: as desc
● Response parameters, types, and descriptions:
    region_id as integer: region identifier
    region_name as string: region name
    state_name as string: state name
    household_income as float: latest household income
    income_rank as integer: rank by income

Route 6:
● Retrieves the average latest home value for each state. This corresponds to Milestone 3 Query 6.
● GET /states/average-home-values
● Request parameters, types, and descriptions: None
● Query parameters (parameters used in the SQL / NoSQL query):
    limit, query as integer: number of rows to return
    sort: as desc
● Response parameters, types, and descriptions:
    state_name as string: state name
    avg_home_value as float: average latest home value for the state

Route 7:
● Retrieves regions ranked by affordability ratio, defined as latest home value divided by latest household income. This corresponds to Milestone 3 Query 7 and is one of the required complex queries.
● GET /rankings/affordability
● Request parameters, types, and descriptions: None
● Query parameters (parameters used in the SQL / NoSQL query):
    limit, query as integer: number of rows to return, default 25
    state_name, query as string: filter by state
    sort, query as string: asc or desc, default desc
    min_ratio, query as float: minimum affordability ratio
    max_ratio, query as float: maximum affordability ratio
● Response parameters, types, and descriptions:
    region_id as integer: region identifier
    region_name as string: region name
    state_name as string: state name
    value as float: latest home value
    household_income as float: latest household income
    affordability_ratio as float: home value divided by household income

Route 8:

● Retrieves regions in the top income quartile that still have high home-price-to-income ratios. This corresponds to Milestone 3 Query 8 and is one of the required complex queries.
● GET /regions/high-income-poor-affordability
● Request parameters, types, and descriptions: None
● Query parameters (parameters used in the SQL / NoSQL query):
    limit, query as integer: maximum number of rows to return
    state_name, query as string: filter by state
    min_ratio, query as float: minimum affordability ratio threshold
● Response parameters, types, and descriptions:
    region_id as integer: region identifier
    region_name as string: region name
    state_name as string: state name
    value as float: latest home value
    household_income as float: latest household income
    affordability_ratio as float: home value divided by household income

Route 9:

● Retrieves home value growth over time for a specific region by comparing each time point with the previous recorded period. This corresponds to Milestone 3 Query 9 and uses a window function.
● /regions/:region_id/home-value-growth
● Request parameters, types, and descriptions:
    region_id, path as integer: target region identifier
● Query parameters (parameters used in the SQL / NoSQL query):
    start_date, query as string: start of date range
    end_date,query as string: end of date range
● Response parameters, types, and descriptions:
    region_id as integer: region identifier
    region_name as string: region name
    state_name as string: state name
    date as string: observation date
    value as float: home value on that date
    prev_value as float: previous period home value
    pct_growth as float: percent change from previous period

Route 10:

● Retrieves regions most similar to a target region based on latest income and latest home value. This corresponds to Milestone 3 Query 10 and is one of the required complex queries.
● GET /regions/:region_id/similar
● Request parameters, types, and descriptions:

region_id, path as integer: target region identifier

- Query parameters (parameters used in the SQL / NoSQL query):
  - limit, query as integer: number of similar regions to return, default 10
  - state_name, query as string: restrict results to one state
- Response parameters, types, and descriptions:
  - region_id as integer: similar region identifier
  - region_name as string: region name
  - state_name as string: state name
  - household_income as float: latest household income
  - value as float: latest home value
  - similarity_score as float: similarity metric based on distance from the target region